assignment 7
sb-prob2910a.problem

Due date: Fri Mar 6 11:59:59 pm 2026 (EST)

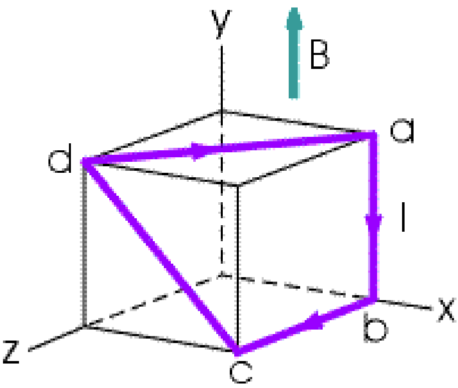
1a. An electron has a velocity of 1.17E+4m/s (in the positive x direction) and an acceleration of 2.39E+12 m/s^2 (in the positive z direction) in uniform electric and magnetic fields. If the electric field has a magnitude of 15.5N/C (in the positive z direction), what is the y component of the magnetic field in the region?

Tries 0/10

1b. What is the z component of the magnetic field in the region?

Tries 0/10

sb-prob2918a.problem
2. In the figure below, the cube is 42.4cm on each edge. Four straight segments of wire - *ab*, *bc*, *cd* and *da* - form a closed loop that carries a current I = 4.92A in the direction shown. A uniform magnetic field of magnitude B = 0.0183T is in the positive y direction.



2a. Determine the magnitude of the magnetic force on segment *ab*.

Tries 0/10

2b. Determine the magnitude of the magnetic force on segment *bc*.

Tries 0/10

2c. Determine the magnitude of the magnetic force on segment *cd*.

Tries 0/10

2d. Determine the magnitude of the magnetic force on segment *da*.

Tries 0/10

sb-prob2926a.problem

A long piece of wire of mass 0.149kg and total length of 3.62m is used to make a square coil with 22 turns. The coil is hinged along a horizontal side, carries a 3.43A current, and is placed in a vertical magnetic field with a magnitude of 0.0136T. Determine the angle that the plane of the coil makes with the vertical when the coil is in equilibrium.

Tries 0/10

Find the torque acting on the coil due to the magnetic force at equilibrium.

Tries 0/10

sb-prob2942a.problem
Singly charged uranium-238 ions are accelerated through a potential difference of 2.00kV and enter a uniform magnetic field of 1.53T directed perpendicular to their velocities. Determine the radius of their circular path.

Tries 0/10

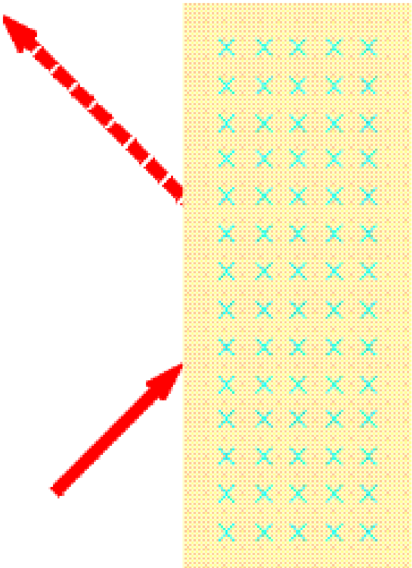
Repeat for uranium-235 ions.

Tries 0/10

sb-prob2944.problem
What is the required radius of a cyclotron designed to accelerate protons to energies of 33.1MeV using a magnetic field of 5.28T?

Tries 0/10

sb-prob2969a.problem
A proton moving in the plane of the page has a kinetic energy of 5.80MeV. It enters a magnetic field of magnitude B = 1.16T directed into the page, moving at an angle of θ= 45.0deg with the straight linear boundary of the field, as shown in the figure below. Calculate the distance x from the point of entry to where the proton leaves the field.



Tries 0/10

Determine the angle between the boundary and the proton's velocity vector as it leaves the field.

Tries 0/10

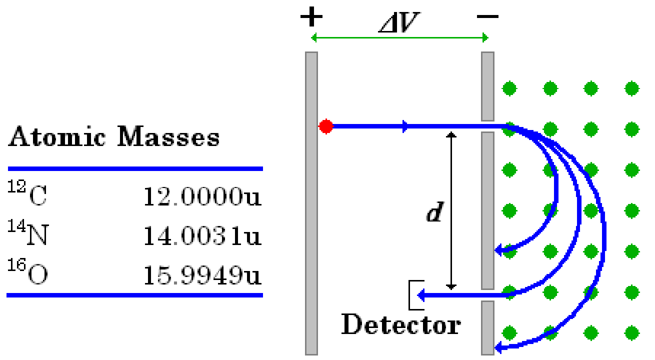
kn-prob3242.problem

A small bar magnet experiences a 0.0422N·m torque when the axis of the magnet is at a 39.3&deg to a 0.427T magnetic field. What is the magnitude of its magnetic dipole moment?

Tries 0/10

kn-prob3270a.problem

The image below shows a *mass spectrometer*, an analytical instrument used to identify the various molecules in a sample by measuring their charge-to-mass ratio e/m. The sample is ionized, the positive ions are accelerated (starting from rest) through a potential difference ΔV, and they then enter a region of uniform magnetic field. The field bends the ions into circular trajectories, but after just half a circle they either strike the wall or pass through a small opening to a detector. As the accelerating voltage is slowly increased, different ions reach the detector and are measured. Typical design values are a magnetic field strength $B = 0.357$ T and a spacing between the entrance and exit holes $d = 8.59$ cm.



What accelerating potential difference ΔV is required to detect $N_2{}^+$?

Tries 0/10

What accelerating potential difference ΔV is required to detect $O_2{}^+$?

Tries 0/10

What accelerating potential difference ΔV is required to detect CO^+?

Tries 0/10

kn-prob3280.problem

The 10 turn loop of wire shown below with l=24.7cm, and w=12.35cm lies in a horizontal plane parallel to a uniform horizontal magnetic field, and carries a 5.45A current. The loop is free to rotate about a nonmagnetic axle through the center. A 89.5g mass hangs from one edge of the loop. What magnetic field strength will prevent the loop from rotating about the axle?

Tries 0/10